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Graduate Complex Analysis

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# A STUDY OF CAUCHY'S INTEGRAL THEOREM

This paper is an analysis of Cauchy's Integral Theorem. In this paper, preliminary examples will be outlined to help the reader follow the proof of Cauchy's Integral Theorem. Applications and additional examples of this theorem will also be presented at the end of this paper. Two important questions that will be addressed are: is the integral value along a path in the complex plane independent of the path, and how do we evaluate contour integrals for contours that enclose singularities of *f(z)*? Below is the theorem in its first form as presented by Knopp [3]:

***First form.*** *Let the function w = f(z) be regular in a simply connected region Ж, and let* $z_0$ *and Z be two (interior) points of Ж. Then the integral*

$$\int_{z_0}^{Z} f(z)dz$$

*has the <u>same value</u> along every path of integration extending from* $z_0$ *to Z and lying entirely within Ж.*

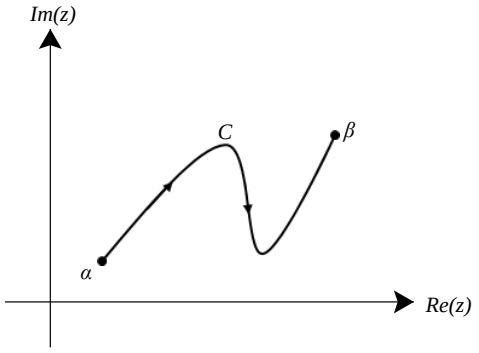
**SECTION 1.** DEFINTION OF LINE INTEGRALS

The integral of a function "*f*" on a complex variable along a given path *C* is called the line integral. The value of this integral varies depending on the path *C*, *f*, and the values of its start point and end point[5]. The following two integral expressions demonstrate how the line integral (these will also be referred to as "Contour Integrals") is written out symbolically:

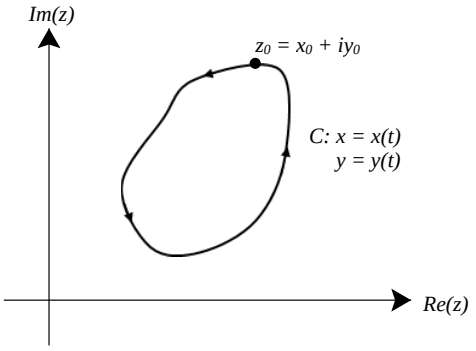
$$\int_C f(z)\,dz \qquad \text{or} \qquad \int_\alpha^\beta f(z)\,dz$$

The figure to the right illustrates graphically the path *C* from α to β. In the complex plane z, α, and β are complex variables of the form "*x + iy*." If the path chosen from α to β is closed then it will be written in the following fashion:

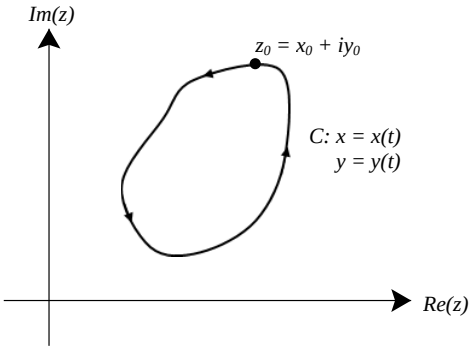
$$\oint_C f(z)\,dz$$




**Example 1:** As an elementary example let us prove $\oint_C dz = 0$, where *C* is every closed contour. First we will evaluate the contour integral from $z_0$ back to $z_0$. Our intuition suggests that this integral should evaluate to the "0" considering that the start and end points are the same location.



The figure to the right shows contour *C* being parameterized as functions *x(t)* and *y(t)*. The complex function is *f(z)* = 1, the integral can be written in the following form:




$$\oint_C 1\, dz = \oint_{z_0}^{z_0} 1\, dz$$

But, we can make the following substitution knowing the parameterization of *C*: *x = x(t)* and *y = y(t)* and then determine the differential *dz*:

$$z = x + iy = x(t) + iy(t)$$

$$\frac{dz}{dt} = \frac{d}{dt}\big(x(t) + iy(t)\big) = x'(t) + iy'(t)$$

$$dz = [x'(t) + iy'(t)]\, dt$$

$$dz = [x_t + iy_t]\, dt$$

The last line above uses the short hand notation for derivatives and will be used in the analysis of this example. We also note that $z_0 = x_0 + iy_0 = x(t_0) + iy(t_0)$ and we will be evaluating this integral from $t_0$ to $t_0$.

$$\oint_{z_0}^{z_0} 1\, dz = \oint_{t_0}^{t_0} 1[x_t + iy_t]\, dt = \oint_{t_0}^{t_0} x_t + iy_t\, dt$$

$$\oint_{z_0}^{z_0} 1\, dz = x(t) + iy(t)]_{t_0}^{t_0} = x(t_0) + iy(t_0) - [x(t_0) + iy(t_0)] = 0$$

**Example 2:** Let us prove $\oint_C z\, dz = 0$. We will make the same parameterization for *C* and follow the same strategy, but the only difference is that *f(z) = z*.

$$\oint_C z\, dz = \oint_{t_0}^{t_0} \underbrace{[x(t) + iy(t)]}_{z}\underbrace{[x_t + iy_t]\, dt}_{dz}$$

Using simple distribution methods we obtain the following integrand:

$$[x(t) + iy(t)][x_t + iy_t] = xx_t - yy_t + i(yx_t + xy_t)$$

The last term $yx_t + xy_t$ is actually an example of the derivative of a product: $yx_t + xy_t = \frac{d}{dt}[x(t)y(t)]$

Let us now re-write the integral formula from above:

$$\oint_{t_0}^{t_0} [x(t) + iy(t)][x_t + iy_t]\, dt = \oint_{t_0}^{t_0} xx_t - yy_t + i\frac{d}{dt}[x(t)y(t)]\, dt$$

$$\oint_{t_0}^{t_0} xx_t\, dt - \oint_{t_0}^{t_0} yy_t\, dt + i\oint_{t_0}^{t_0} \frac{d}{dt}[x(t)y(t)]dt \qquad (A)$$

The last term is an integral of a derivative, this will leave us with the product *x(t)y(t)*. The first two integrals on the other hand will require some other techniques. If we let u = *x(t)* then du = $x_t\, dt$ and if we make a similar substitution for the second integral we will obtain the following results:

$$\oint xx_t\, dt = \oint u\, du \quad \text{where } u = x(t) \text{ and } du = x_t\, dt$$

$$\boxed{\oint u\,du = \frac{1}{2}u^2 + c = \frac{1}{2}[x(t)]^2 + c}$$

$$\oint yy_t\,dt = \oint v\,dv \quad \text{where } v = y(t) \text{ and } dv = y_t\,dt$$

$$\boxed{\oint v\,dv = \frac{1}{2}v^2 + c = \frac{1}{2}[y(t)]^2 + c}$$

If we make the replacements into the integral equation from line (A) we obtain the following:

$$\oint_{t_0}^{t_0} xx_t\,dt - \oint_{t_0}^{t_0} yy_t\,dt + i\oint_{t_0}^{t_0} \frac{d}{dt}[x(t)y(t)]\,dt = \frac{1}{2}[x(t)]^2 - \frac{1}{2}[y(t)]^2 + x(t)y(t)]_{t_0}^{t_0}$$

$$\left(\frac{1}{2}[x(t_0)]^2 - \frac{1}{2}[y(t_0)]^2 + x(t_0)y(t_0)\right) - \left(\frac{1}{2}[x(t_0)]^2 - \frac{1}{2}[y(t_0)]^2 + x(t_0)y(t_0)\right) = 0$$

**Example 3:** Let us prove $\oint_C z^2\,dz = 0$. We will make the same parameterization for *C* and follow the same strategy as the previous two examples, but the only difference is that *f(z)* = $z^2$. First we need to determine *f(z)* in terms of *t*:

$$f(z) = z^2 = (x + iy)^2 = x^2 - y^2 + i2xy \quad \text{where } x = x(t) \text{ and } y = y(t)$$

As before if *z* = *x* + *iy* then *dz* = $x_t + iy_t\,dt$. Next we will determine the integrand that needs to be evaluated:

$$f(z)\,dz = [x^2 - y^2 + i2xy][x_t + iy_t]dt$$

$$f(z)\,dz = [x^2x_t - y^2x_t - 2xyy_t + i(2xx_ty + x^2y_t - y^2y_t)]\,dt$$

These two terms can be obtained using the chain rule and product rule:

$$\frac{d}{dt}[y^2x] = y^2x_t + 2xyy_t$$

These two terms can be obtained using the chain rule and product rule:

$$\frac{d}{dt}[x^2y] = x^2y_t + 2xx_ty$$

Using these analysis techniques the integrand is obtained to be the following:

$$f(z)\,dz = \left[x^2x_t - \frac{d}{dt}[y^2x] + i\left(\frac{d}{dt}[x^2y] - y^2y_t\right)\right]\,dt$$

The following integral needs to be evaluated:

$$\oint f(z)\,dz = \oint x^2x_t\,dt - \oint \frac{d}{dt}[y^2x]\,dt + i\oint \frac{d}{dt}[x^2y]\,dt - i\oint y^2y_t\,dt \qquad (B)$$

Please take note that the first and last integral can be determined using simple substitution:

$$\oint x^2x_t\,dt = \oint u^2\,du \quad \text{where } u = x(t) \text{ and } du = x_t\,dt$$

$$\boxed{\oint u^2\,du = \frac{1}{3}u^3 + c = \frac{1}{3}[x(t)]^3 + c}$$

$$\oint y^2y_t\,dt = \oint v^2\,dv \quad \text{where } v = y(t) \text{ and } dv = y_t\,dt$$

$$\boxed{\oint v^2\,dv = \frac{1}{3}v^3 + c = \frac{1}{3}[y(t)]^3 + c}$$

If we make the above replacement into line (B) we can evaluate:

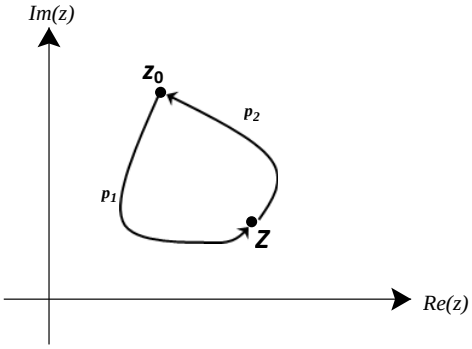
$$\frac{1}{3}[x(t)]^3 - [y(t)]^2x(t) + i[x(t)]^2y(t) - i\frac{1}{3}[y(t)]^3\Big]_{t_0}^{t_0}$$

$$= \left(\frac{1}{3}[x(t_0)]^3 - [y(t_0)]^2x(t_0) + i[x(t_0)]^2y(t_0) - i\frac{1}{3}[y(t_0)]^3\right)$$
$$-\left(\frac{1}{3}[x(t_0)]^3 - [y(t_0)]^2x(t_0) + i[x(t_0)]^2y(t_0) - i\frac{1}{3}[y(t_0)]^3\right) = 0$$

**SECTION 2.** PROOF OF CAUCHY'S INTEGRAL THEOREM

Looking back at the first form of Cauchy's Integral Theorem one can conclude that if the value of the integral is independent of the path taken from $z_0$ to Z then the line integrals of the two paths $p_1$ and $p_2$ are equal [3]:

$$\int_{p_1} f(z)\,dz = \int_{p_2} f(z)\,dz \quad \leftrightarrow \quad \int_{p_1} f(z)\,dz - \int_{p_2} f(z)\,dz = \oint_C f(z)\,dz = 0$$




Let $C: p_1 \cup -p_2$ then we just need to prove that the closed contour integral along $C$ is equal to 0.

This proof will be broken into three parts, Triangular Paths, Polygonal Paths, and then generalized to arbitrary paths [3]. The illustration below gives a sense of how this proof will progress:

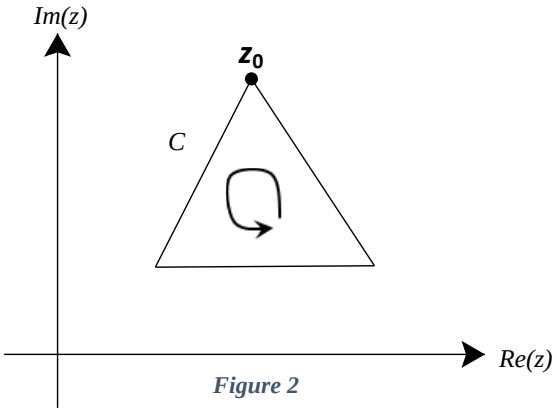



*Figure 2*

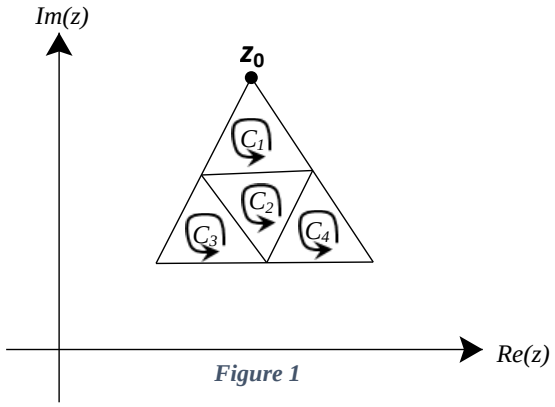



*Figure 1*

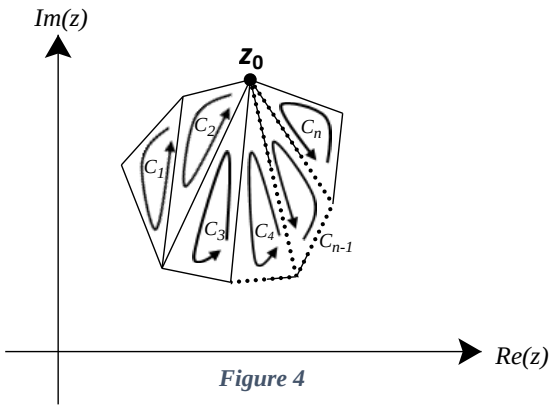



*Figure 4*

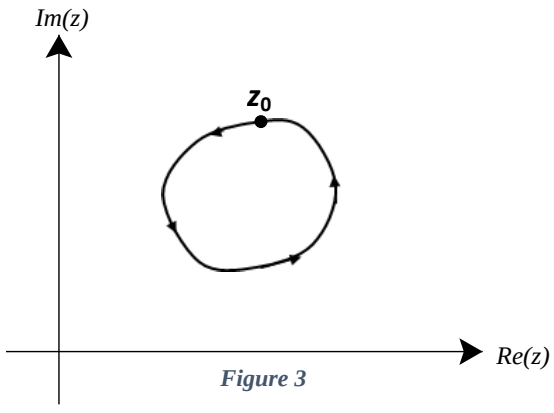
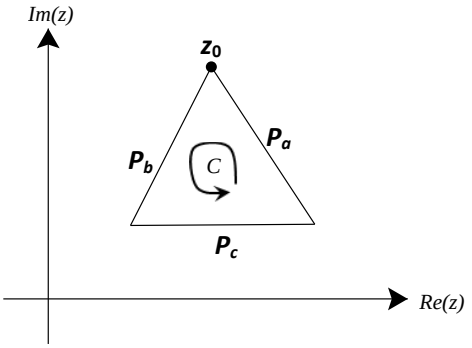



*Figure 3*

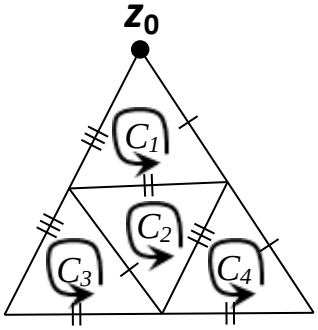
**Part 1:** First, we will consider a triangular path starting and ending at $z_0$. We can write the contour integral for the path in the figure to the right as the following:




$$\oint_C f(z)dz = \oint_{P_a+P_b+P_c} f(z)dz$$

But this only relates *C* to the singular paths $P_a, P_b, P_c$. Let us consider figure 2 above and subdivide the larger triangle into four smaller triangles by drawing segments in the interior of the large triangle that run parallel to the sides of the larger triangle.

If those parallel segments are drawn such that they connect the three midpoints of the larger triangle then we have the following:




$$\oint_C f(z)dz = \oint_{C_1} f(z)\,dz + \oint_{C_2} f(z)\,dz + \oint_{C_3} f(z)\,dz + \oint_{C_4} f(z)\,dz$$

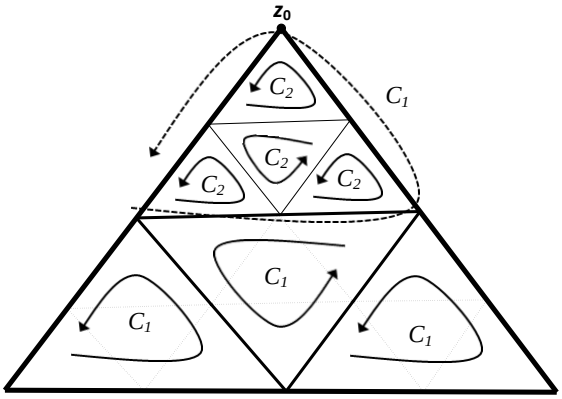
*Recall from elementary geometry, the four triangles are congruent to each other and similar in size to the larger triangular path *C* [1].

Hence, $\Delta C_1 \cong \Delta C_2 \cong \Delta C_3 \cong \Delta C_4$ and, if we use $C_1$ as the principle contour then the magnitude of the above integral is expressed in the following way:

$$\left|\oint_C f(z)dz\right| \le 4\left|\oint_{C_1} f(z)\,dz\right| \qquad (C)$$

Notice, that this is an inequality relationship because the contour integral over the larger triangle *C* does not take into account the interior paths formed by the parallel lines. The right hand side is a simple product of 4 and the contour integral $C_1$ and this includes the interior paths formed by the parallel segments. We are using the magnitudes of the contour integrals to prove that they converge to 0, and if the magnitudes converge then so do the integrals.

Subdivide $C_1$ using parallel segments to obtain a nested set of smaller similar triangles. With this further subdivision of $C_1$ we can write another integral relationship. Please see the diagram to the right [4]:




$$\left|\oint_{C_1} f(z)dz\right| \le 4\left|\oint_{C_2} f(z)\,dz\right|$$

Substitute the above relationship into line (C) to obtain the following:

$$\left|\oint_C f(z)dz\right| \le 4\left|4\left|\oint_{C_2} f(z)\,dz\right|\right|$$

$$\left|\oint_C f(z)dz\right| \le 4^2\left|\oint_{C_2} f(z)\,dz\right|$$

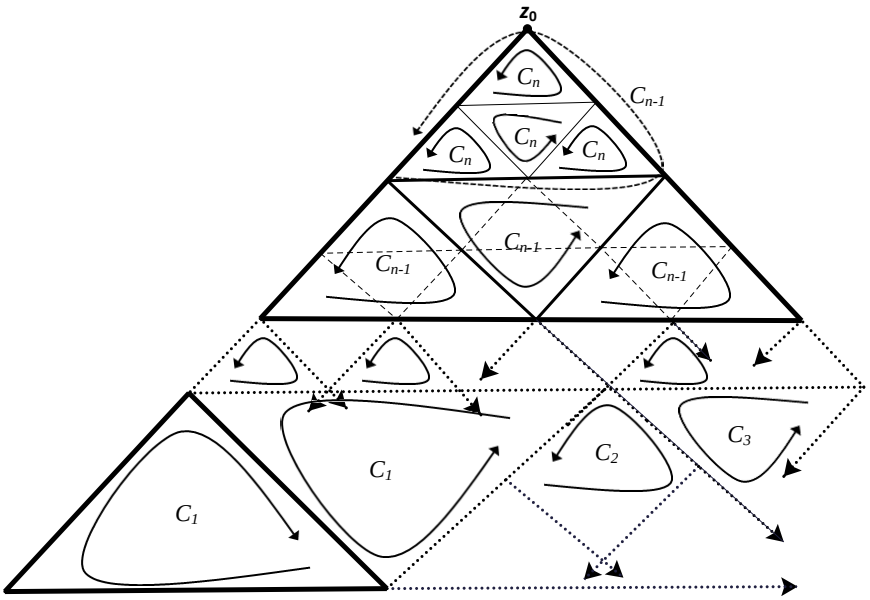
Clearly, there is a pattern with the subdivision of the triangular paths. Further subdivision of the larger triangle *n* times will establish the following (notice, this pattern of self-similarity resembles the Sierpinski Triangle):

$$\left|\oint_C f(z)dz\right| \le 4^n\left|\oint_{C_n} f(z)\,dz\right| \qquad (**)$$

Upon examination of this relationship, our intuition tells us that this can be taken to a limit. Let's take a step back and restate the definition of a derivative at a point in the complex plane:

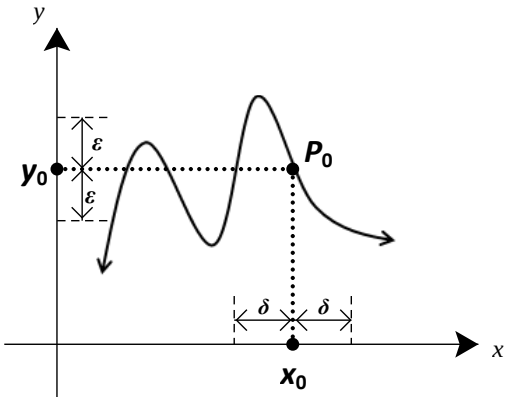
$$f'(z_0) = \lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0} \qquad (D)$$

If we imagine that $z$ gets extremely close to $z_0$, then there will be an *error* which we will refer symbolically as $\varepsilon$. What does the limit look like graphically? Below are diagrams showing the limit being taken in the real-plane and the complex plane.




In the real plane the epsilon-delta definition of the derivative can be proved using the following inequalities

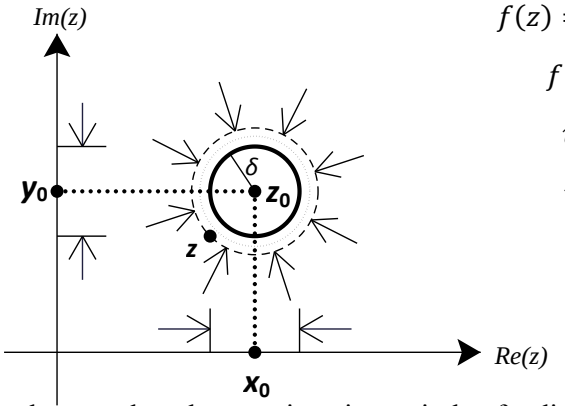
$$\left| \frac{f(x) - f(x_0)}{x - x_0} - f'(x_0) \right| < \varepsilon \qquad \text{with} \quad 0 < |x - x_0| < \delta$$




The goal is to choose $\varepsilon$ as small as possible to make δ as small as possible; additionally, using these inequalities we can express δ as a function for $\varepsilon$:

$$\delta = \gamma(\varepsilon)$$

Below are the diagrams for finding the limit in the complex-plane, we need to treat *f(z)* as a function in the u-v plane that corresponds to the point *z* in the complex plane:

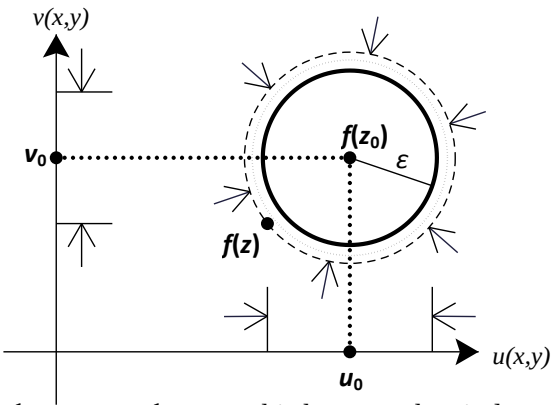



$$f(z) = u(x,y) + iv(x,y)$$

$$f(z_0) = u_0 + iv_0$$

$$u_0 = u(x_0, y_0)$$

$$v_0 = v(x_0, y_0)$$




In the complex plane we imagine a circle of radius δ shrinking closer and closer towards $z_0$, as this happens the circle surrounding $f(z_0)$ has radius $\varepsilon$, approaches zero. Therefore, we can write inequalities to determine a relationship between δ and $\varepsilon$, such that, choose $\varepsilon$ so small that δ becomes insignificant. Remove the limit from (D) and let *z* approach $z_0$, the following inequality statements will result [2][3]:

$$\left| \frac{f(z) - f(z_0)}{z - z_0} - f'(z_0) \right| < \varepsilon \quad (1) \qquad \text{with} \quad 0 < |z - z_0| < \delta \qquad (2)$$

Take (1) and multiply both sides of the inequality by $|z - z_0|$:

$$|f(z) - f(z_0) - f'(z_0)(z - z_0)| < \varepsilon |z - z_0| \qquad (3)$$

Using (D), subtract $f'(z_0)$ from both sides to obtain the following equation:

$$\left(\lim_{z\to z_0}\frac{f(z)-f(z_0)}{z-z_0}\right)-f'(z_0)\sim 0\ (\text{for } z \text{ really close to } z_0)=\sigma$$

If $z$ is chosen much closer to $z_0$ then σ will be essentially 0 if not it will be extremely close to it and its magnitude is less than $\varepsilon$; therefore $|\sigma|<\varepsilon$. If we write (3) in equality form, we can use the σ from above. Then, solve for *f(z)*.

$$f(z)-f(z_0)-\ f'(z_0)(z-z_0)=\sigma\cdot(z-z_0)\quad\leftrightarrow\quad f(z)=f(z_0)+\ f'(z_0)(z-z_0)+\sigma\cdot(z-z_0)$$

$$f(z)=f(z_0)+\ zf'(z_0)-z_0f'(z_0)+\sigma\cdot(z-z_0)\qquad(4)$$

We will take the contour integral on both sides of (4) and the following will result:

$$\oint_{C_n}f(z)\,dz=\oint_{C_n}f(z_0)+\ zf'(z_0)-z_0f'(z_0)+\sigma\cdot(z-z_0)\,dz$$

$$=\oint_{C_n}f(z_0)\ dz+\oint_{C_n}zf'(z_0)\ dz-\oint_{C_n}z_0f'(z_0)\,dz+\oint_{C_n}\sigma\cdot(z-z_0)\,dz$$

$$=f(z_0)\underbrace{\oint_{C_n}1\ dz}_{0}+\ f'(z_0)\underbrace{\oint_{C_n}z\ dz}_{0}-z_0f'(z_0)\underbrace{\oint_{C_n}1\,dz}_{0}+\oint_{C_n}\sigma\cdot(z-z_0)\,dz$$

Using the results of the first two example problems presented at the beginning of this paper, the first three integrals will evaluate to 0.

$$\oint_{C_n}f(z)\,dz=0+0+0+\oint_{C_n}\sigma\cdot(z-z_0)\ dz$$

$$\oint_{C_n}f(z)\,dz=\sigma\oint_{C_n}(z-z_0)\ dz$$

For our purposes we are using the magnitudes of the integrals, so we will take the magnitude operator on both sides of the equality:

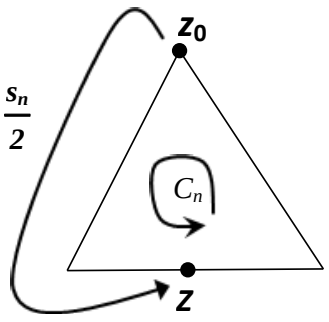
$$\left|\oint_{C_n}f(z)\,dz\right|=|\sigma|\left|\oint_{C_n}(z-z_0)\ dz\right|$$

Again, we are using the magnitudes to prove absolute convergence because if the magnitudes converge to 0, then the contour integrals converge to 0. But let us remember that $|\sigma|<\varepsilon$ and distributing the magnitude to the integrand will form an inequality relationship:

$$\left|\oint_{C_n}f(z)\,dz\right|=|\sigma|\left|\oint_{C_n}(z-z_0)\ dz\right|<\ \varepsilon\oint_{C_n}|(z-z_0)|\,|dz|\qquad(5)$$

What does the integral on the right side actually evaluate to? Remember that this inequality holds if we imagine that $z$ and $z_0$ are furthest apart and both points will be at the maximum 'distance' along the triangular path. The 'distance' is distance along the path connecting $z_0$ to *Z*. The differential *dz* has magnitude equal to the perimeter $s_n$ of the triangle because we are integrating along all three sides of the triangle [3]. The figure to the right illustrates these quantities.

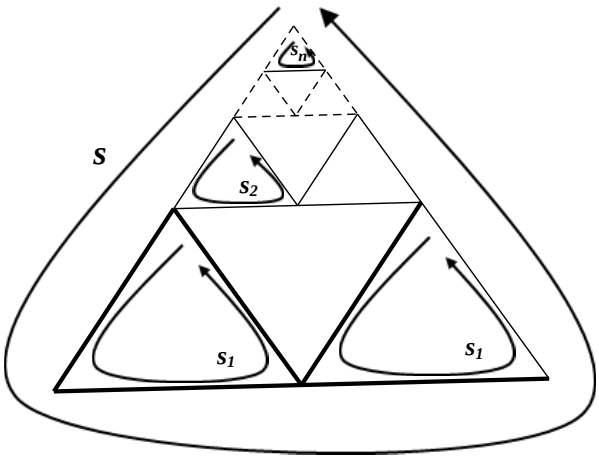
$$|(z-z_0)|=\max_{z\to z_0}C_n=\frac{Perimeter}{2}=\frac{s_n}{2}$$




We can now use these quantities to re-express (5):

$$\left|\oint_{C_n}f(z)\,dz\right|<\ \varepsilon\left(\frac{s_n}{2}\right)s_n=\varepsilon\frac{s_n^2}{2}\qquad(6)$$

Take notice that we still need to relate $s_n$ to the whole perimeter of the triangular path *s*. If we refer back to our high school geometry, the similar triangles nested in the larger triangle referred to before, have the following property. The perimeter of a triangle is equal to half the perimeter of the larger triangle preceding it. Refer to the illustration at the right.




$$s_1 = \frac{s}{2}\,, s_2 = \frac{s_1}{2} = \frac{s}{2\cdot 2}\,, s_3 = \frac{s_2}{2} = \frac{s}{2\cdot 2\cdot 2}$$

If we continue this pattern we can express the perimeter for the nth subdivision as:

$$s_n = \frac{s}{2^n}$$

Let us make the substitution into (6) and we get the following inequality:

$$\left|\oint_{C_n} f(z)\,dz\right| < \varepsilon\left(\frac{s_n}{2}\right)s_n = \varepsilon\frac{s_n^2}{2} = \varepsilon\left(\frac{1}{2}\right)\left(\frac{s}{2^n}\right)^2 = \varepsilon\left(\frac{s^2}{2\cdot 4^n}\right)$$

$$\left|\oint_{C_n} f(z)\,dz\right| < \varepsilon\left(\frac{s^2}{2\cdot 4^n}\right)$$

Use (**) and the above inequality to establish a relationship between the contour integral of the whole triangular path and the continuous subdivision of the smaller triangles:

$$\left|\oint_{C} f(z)dz\right| \le 4^n\left|\oint_{C_n} f(z)\,dz\right| \le 4^n\cdot\varepsilon\left(\frac{s^2}{2\cdot 4^n}\right)$$

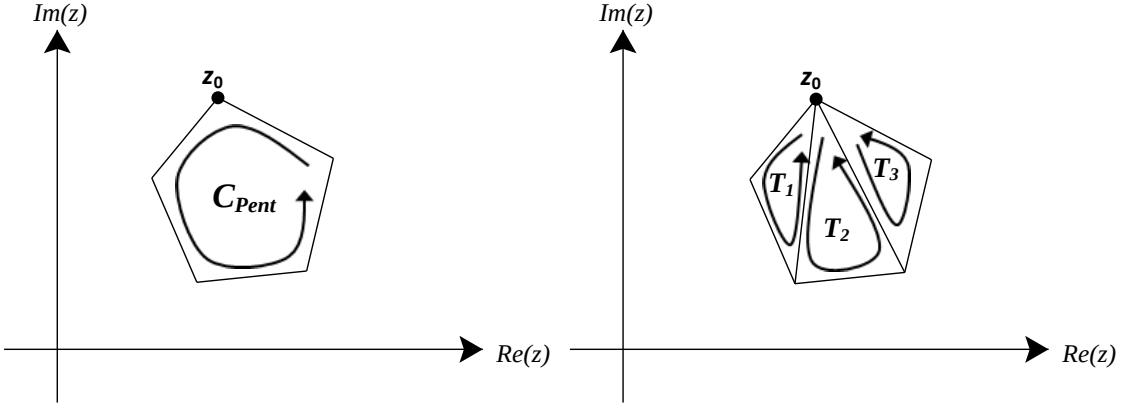
$$\left|\oint_{C} f(z)dz\right| \le 4^n\left|\oint_{C_n} f(z)\,dz\right| \le \varepsilon\left(\frac{s^2}{2}\right)$$

If $\varepsilon$ = 0, then..

$$\left|\oint_{C} f(z)dz\right| \le 4^n\left|\oint_{C_n} f(z)\,dz\right| \le 0$$

If we let $\varepsilon$ get closer and closer to 0 the entire inequality becomes zero; therefore, the magnitude of the integral converges to 0. This means that the contour integral along a closed path is 0; hence, the value of the integral for the path taken between two points in the complex plane is independent of the path chosen [3].

**Part 2:** Suppose that *C* is the closed polygonal path, as an example let the path be the pentagon $C_{Pent}$ shown below. Subdivide this pentagonal path into three triangles by drawing every diagonal leaving point $z_0$.




Using the two diagrams above, we can write a relationship regarding the contour integrals about the pentagonal path:

$$\oint_{C_{Pent}} f(z)\,dz = \oint_{T_1} f(z)\,dz + \oint_{T_2} f(z)\,dz + \oint_{T_3} f(z)\,dz$$

If we look back at Part 1 of the proof we concluded that the contour integral taken along a closed triangular path is equal to 0, therefore:

$$\oint_{C_{Pent}} f(z)\,dz = 0 + 0 + 0 = 0$$

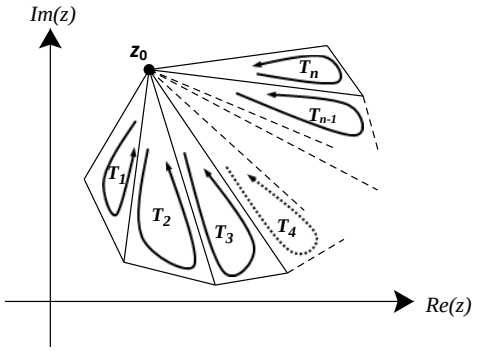
$$\oint_{C_{Pent}} f(z)\,dz = 0$$

Imagine a polygon with n vertices then we have the following diagram and corresponding contour integral.

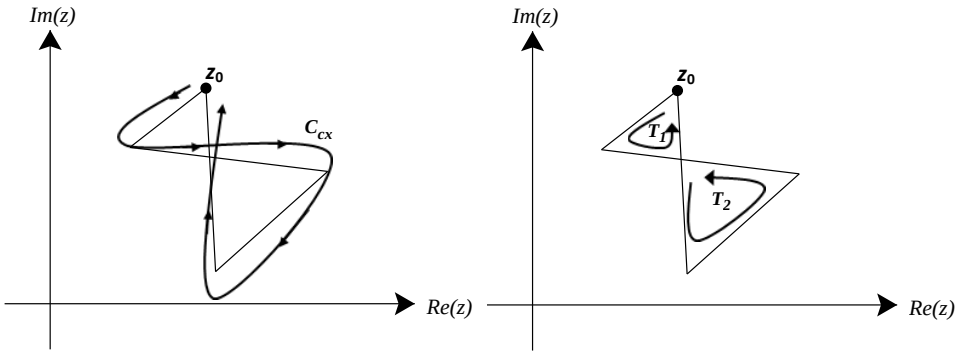
$$\oint_{C_{n-gon}} f(z)\,dz = \oint_{T_1} f(z)\,dz + \oint_{T_2} f(z)\,dz + \oint_{T_3} f(z)\,dz + \cdots + \oint_{T_n} f(z)\,dz$$

$$\oint_{C_{n-gon}} f(z)\,dz = 0 + 0 + 0 + \cdots + 0 = 0$$

$$\oint_{C_{n-gon}} f(z)\,dz = 0$$




Again, from Part 1 of the proof, each contour integral along a triangular path evaluates to 0. This means that the sum of all the triangular paths will sum to 0.We have yet to consider self-intersecting polygons, the above part of the proof involved simple polygons.




Illustrated above is a self-intersecting polygon in the complex plane, if we divide the contour into two triangles we can write the following contour integrals. Please take note that $T_2$ runs in an opposite direction as compared to $C_{cx}$, this means the negative of $T_2$ will be used.

$$\oint_{C_{cx}} f(z)\,dz = \oint_{T_1} f(z)\,dz - \oint_{T_2} f(z)\,dz$$

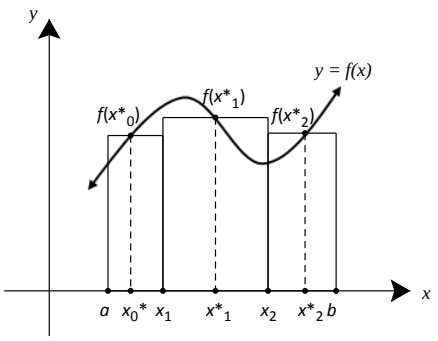
Again, the integrals along the triangular contours evaluate to 0.

$$\oint_{C_{cx}} f(z)\,dz = 0 - 0$$

$$\oint_{C_{cx}} f(z)\,dz = 0$$

We have demonstrated that a closed polygonal path, simple or self-intersecting, still yields the same value for the contour integral.

**Part 3:** The last part of the proof will employ epsilon, delta definitions of limits and summations. First recall that in the real plane the integral is defined as the following [5]:

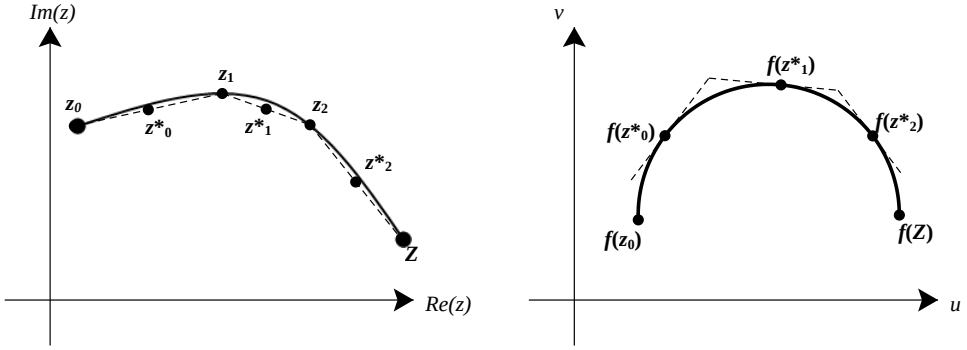



$$\int_a^b f(x)\,dx = \lim_{n\to\infty}\sum_{p=0}^{n}\left(x_p - x_{p-1}\right)f\left(x_p^*\right) = I$$

Where $x_{p-1} < x_p^* < x_p$ and $x_p \in [a, b]$. Imagine the rectangles are getting progressively thinner as we continue to decrease the subinterval size and add another rectangle. This same type of integral can be applied in the complex plane on complex variables, *z* [3].

$$\mathcal{J} = \int_C f(z)\,dz = \int_{z_0}^{Z} f(z)\,dz = \lim_{n\to\infty}\sum_{p=0}^{n}\left(z_p - z_{p-1}\right)f\left(z_p^*\right) \qquad (E)$$

The only difference here is that the integral on a complex function is over a path *C* where the differential *dz* are the differences of the subdivided sections of the path *C*.




But the same will occur in the complex plane as did in the real plane. As $n \to \infty$ and we increase the subdivisions along, then the difference between the straight line paths *C* and the path formed by *f(z)*, becomes smaller and smaller. The error between the straight line paths and the curved paths approaches 0. Equation (E) involves a limit, remove the limit to get the following:

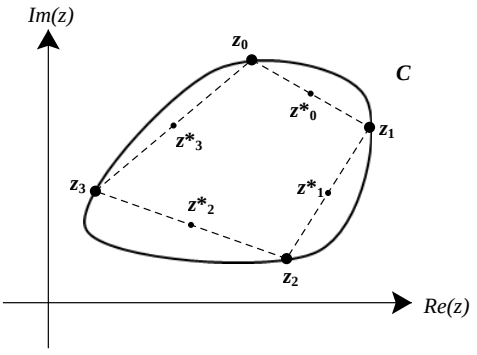
$$\mathcal{J}_n = \sum_{p=0}^{n}\left(z_p - z_{p-1}\right)f\left(z_p^*\right)$$

$$\mathcal{J}_n < \mathcal{J} \quad \Rightarrow \quad \mathcal{J} - \mathcal{J}_n \sim 0 \text{ as } n \to \infty$$

As *n* approaches ∞, the difference in $\mathcal{J}_n$ and $\mathcal{J}$ approaches 0, if we choose an error $'\varepsilon'$ to represent the magnitude of the difference we obtain:

$$|\mathcal{J} - \mathcal{J}_n| < \varepsilon \qquad (F)$$

Now we need to consider a closed smooth contour being approximated by a polygon with n-vertices. It needs to be shown that a closed polygonal path can be continuously subdivided to approximate the smooth contour. As was shown in Part 2, the integral over a closed polygon is 0. We will approximate the contour integral *C* by using the polygonal path:

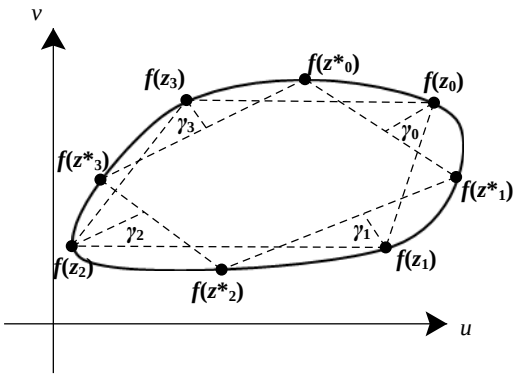



$$\mathcal{J} = \oint_C f(z)\,dz \approx \oint_{poly} f(z)\,dz$$

$$\oint_{poly} f(z)\,dz = \int_{\overline{z_0 z_1}} f(z)\,dz + \int_{\overline{z_1 z_2}} f(z)\,dz + \cdots + \int_{\overline{z_n z_0}} f(z)\,dz$$

$$\mathcal{J}_{Poly} = \oint_{poly} f(z)\,dz = \sum_{p=1}^{n} \int_{\overline{z_{p-1}z_p}} f(z)\,dz \qquad (7)$$

Let us consider *f(z)* in the u-v plane shown to the right, the difference γ can be used to express *f(z)*. The quantity γ is the error between $f(z_p^*)$ and $f(z_p)$, as we decrease the subinterval size and increase the number of subintervals γ will become negligible.




$$f(z_p) - f(z_p^*) = \gamma_p \qquad \leftrightarrow \qquad f(z_p) = f(z_p^*) + \gamma_p$$

Where $|\gamma_p| \ll \varepsilon$, as the number of subdivisions increases. Substitute this into (7) we obtain the following:

$$\mathcal{J}_{Poly} = \oint_{poly} f(z)\,dz = \sum_{p=1}^{n} \int_{\overline{z_{p-1}z_p}} \left(f(z_p^*) + \gamma_p\right) dz$$

Distribute the summation and the integral to re-write $\mathcal{J}_{Poly}$.

$$\sum_{p=1}^{n} \int_{\overline{z_{p-1}z_p}} \left(f(z_p^*) + \gamma_p\right) dz = \sum_{p=1}^{n} \int_{\overline{z_{p-1}z_p}} f(z_p^*)\,dz + \sum_{p=1}^{n} \int_{\overline{z_{p-1}z_p}} \gamma_p\ dz$$

$$\mathcal{J}_{Poly} = \sum_{p=1}^{n} f(z_p^*) \int_{\overline{z_{p-1}z_p}} dz + \sum_{p=1}^{n} \int_{\overline{z_{p-1}z_p}} \gamma_p\ dz \qquad (8)$$

The integral $\int_{\overline{z_{p-1}z_p}} dz = \int_{z_{p-1}}^{z_p} dz = (z_p - z_{p-1})$, we can make this a substitution into (8):

$$\mathcal{J}_{Poly} = \sum_{p=1}^{n} f(z_p^*)(z_p - z_{p-1}) + \sum_{p=1}^{n} \int_{\overline{z_{p-1}z_p}} \gamma_p\ dz$$

Take notice that the first summation above is just the definition of $\mathcal{J}_n$.

$$\mathcal{J}_{Poly} = \mathcal{J}_n + \sum_{p=1}^{n} \int_{\overline{z_{p-1}z_p}} \gamma_p\ dz \quad \rightarrow \mathcal{J}_{Poly} - \mathcal{J}_n = \sum_{p=1}^{n} \int_{\overline{z_{p-1}z_p}} \gamma_p\ dz$$

$$\left|\mathcal{J}_{Poly} - \mathcal{J}_n\right| = \left|\sum_{p=1}^{n} \int_{\overline{z_{p-1}z_p}} \gamma_p\ dz\right| \leq \sum_{p=1}^{n} \int_{\overline{z_{p-1}z_p}} |\gamma_p|\ |dz|$$

$$\left|\mathcal{J}_{Poly} - \mathcal{J}_n\right| \leq \sum_{p=1}^{n} \int_{\overline{z_{p-1}z_p}} |\gamma_p|\ |dz| \ll \sum_{p=1}^{n} \varepsilon \cdot (z_p - z_{p-1}) \ll \varepsilon \qquad (9)$$

If we use the magnitude of the difference $\mathcal{J}_{Poly} - \mathcal{J}_n$ this will be much less than the error $\varepsilon$ as we choose smaller and smaller subdivisions of the contour *C* to approximate. In fact, line (9) contains the summation of $(z_p - z_{p-1})$. Therefore, as $z_p \rightarrow z_{p-1}$ the difference $(z_p - z_{p-1}) \rightarrow 0$. The last summation in (9) closes more rapidly to 0 and will always be much less than the $\varepsilon$ we chose.

Observe the following magnitude:

$$\left|\mathcal{J} - \mathcal{J}_{Poly}\right| \leq ??$$

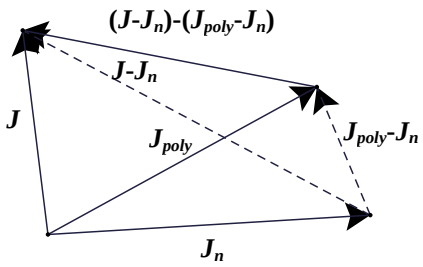



Figure 1

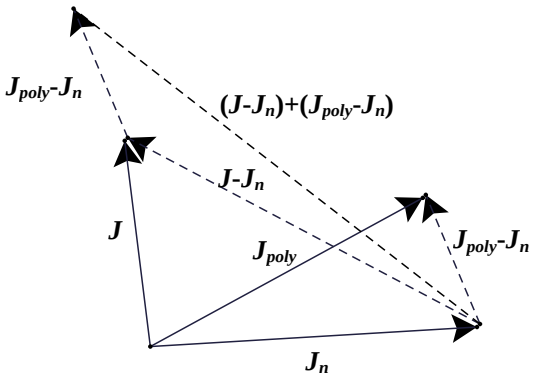



Figure 2

How will this compare to $\varepsilon$? If we expand the above unknown inequality using $\mathcal{J}_n$ we establish the following:

$$\left|\mathcal{J} - \mathcal{J}_{Poly}\right| = \left|(\mathcal{J} - \mathcal{J}_n) - \left(\mathcal{J}_{Poly} - \mathcal{J}_n\right)\right| \quad (10)$$

This can be interpreted in vector notation, please see figure 1. Distribute the magnitude operator in (10) and describe this as vectors in a plane, we obtain the following (figure 2):

$$|(\mathcal{J} - \mathcal{J}_n)| + \left|\left(\mathcal{J}_{Poly} - \mathcal{J}_n\right)\right| \quad (11)$$

From a geometric interpretation (11) is always greater than (10):

$$\left|(\mathcal{J} - \mathcal{J}_n) - \left(\mathcal{J}_{Poly} - \mathcal{J}_n\right)\right| \leq |(\mathcal{J} - \mathcal{J}_n)| + \left|\left(\mathcal{J}_{Poly} - \mathcal{J}_n\right)\right|$$

Keep in mind the inequalities from (F) and (9), we can establish the following:

$$\left|(\mathcal{J} - \mathcal{J}_n) - \left(\mathcal{J}_{Poly} - \mathcal{J}_n\right)\right| \leq |(\mathcal{J} - \mathcal{J}_n)| + \left|\left(\mathcal{J}_{Poly} - \mathcal{J}_n\right)\right| \ll \varepsilon + \varepsilon = 2\varepsilon$$

Therefore,

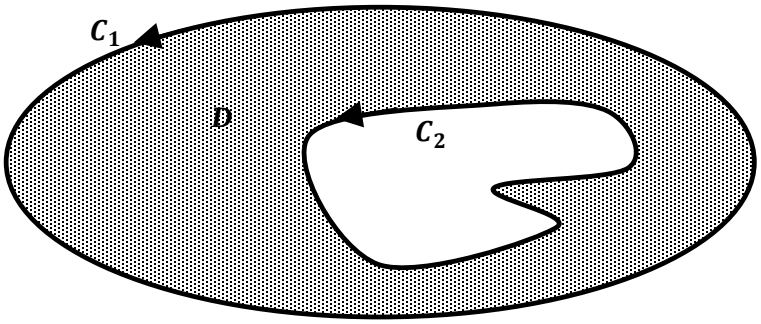
$$\left|\mathcal{J} - \mathcal{J}_{Poly}\right| \ll 2\varepsilon$$

From this statement we can make $\varepsilon$ as arbitrarily small as possible when $\mathcal{J} \to \mathcal{J}_{Poly}$, and from Part 2 of the proof we demonstrated the integral over a polygonal path equals 0, hence if $\mathcal{J}_{Poly} = 0$ and let $\varepsilon \to 0$, then $\mathcal{J} \to 0$.

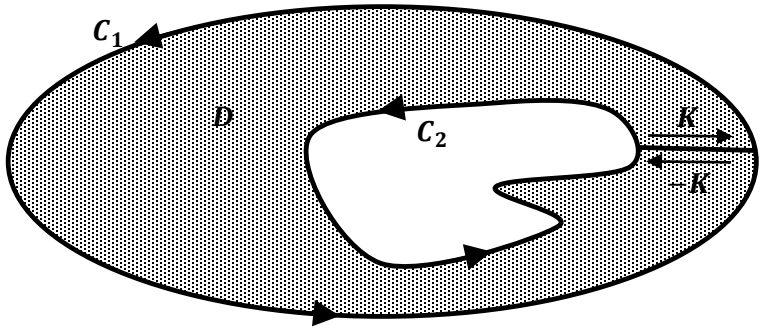
$$\mathcal{J} = \oint_C f(z)\, dz = 0$$

**SECTION 3.** CAUCHY'S INTEGRAL THEOREM FOR MULTIPLY CONNECTED DOMAINS

Suppose contour $C_1$ encompasses $C_2$ as illustrated below and the shaded region is the domain *D*. We will consider counter clockwise the *positive* orientation and clockwise the *negative* orientation [3].



If we make a cut through the domain *D* we can form one continuous closed contour integral and use Cauchy's Integral Theorem:



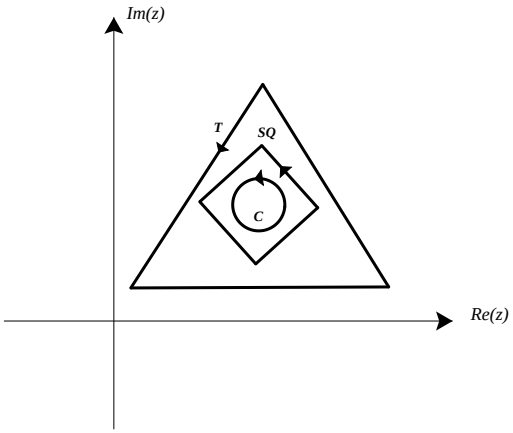

$$\oint_{C_1} f(z)\,dz + \oint_{-K} f(z)\,dz - \oint_{C_2} f(z)\,dz + \oint_{K} f(z)\,dz = 0$$

However, $\oint_{-K} f(z)\,dz = -\oint_{K} f(z)\,dz$, substitute this into the equation above.

$$\oint_{C_1} f(z)\,dz - \oint_{K} f(z)\,dz - \oint_{C_2} f(z)\,dz + \oint_{K} f(z)\,dz = 0$$

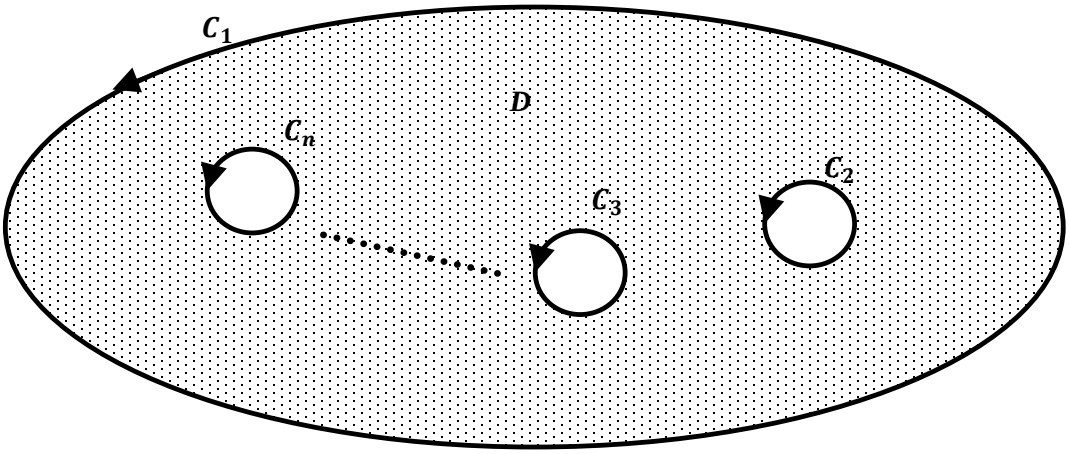
$$\oint_{C_1} f(z)\,dz - \oint_{C_2} f(z)\,dz = 0 \quad \Leftrightarrow \quad \oint_{C_1} f(z)\,dz = \oint_{C_2} f(z)\,dz \quad (***)$$

This last statement means that: if we have a multiply connected region, the integral evaluated over the outer contour is equal to the integral evaluated over the contour it encompasses. This also establishes a very profound thought, if we enclose a region with a contour and we have a contour completely within that contour, then the contour integrals will be equal. Hence, we could have in the complex plane a circle inside a square that is inside a triangle and the contour integrals would be equal to each other!

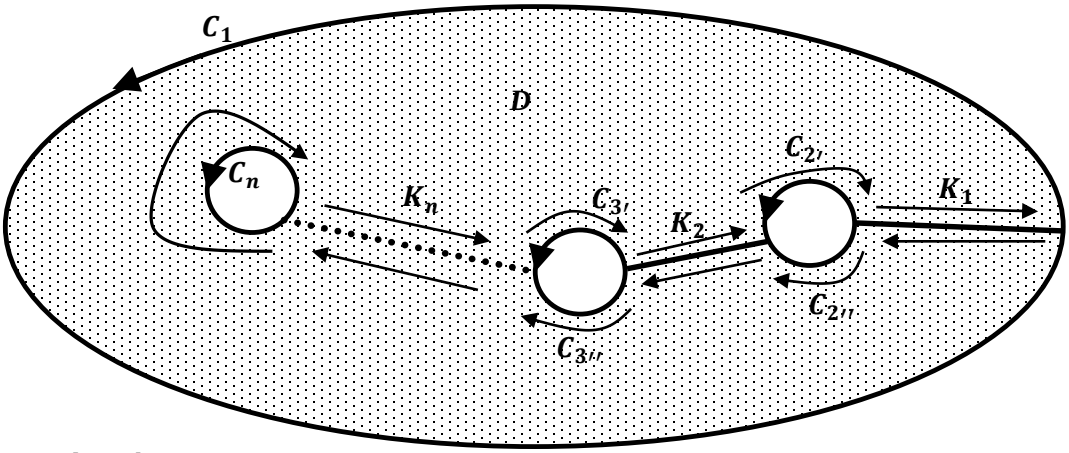



$$\oint_{T} f(z)\,dz = \oint_{SQ} f(z)\,dz = \oint_{C} f(z)\,dz$$

What if there are n-closed paths within $C_1$ as shown below, where $C_1$ is in a positive orientation?



We will treat the problem same as before, by first making a cut that links all of the enclosed paths to each other and to the larger path $C_1$. Please see the next figure.




The following integral can be written:

$$\oint_{C_1} + \int_{-K_1} - \int_{C_{2}''} + \int_{-K_2} - \int_{C_{3}''} + \ldots + \int_{-K_n} - \oint_{C_n} + \int_{K_n} \ldots - \int_{C_{3}'} + \int_{K_2} - \int_{C_{2}'} + \int_{K_1} = 0$$

The *K* paths cancel out and $C_{p}''$ and $C_{p}'$ combine to $C_p$:

$$\oint_{C_1} - \oint_{C_2} - \oint_{C_3} \ldots - \oint_{C_n} = 0$$

Now move integral terms to the other side of the equal sign:

$$\oint_{C_1} = \oint_{C_2} + \oint_{C_3} + \cdots + \oint_{C_n}$$

Therefore, if a contour $C_0$ encloses n-contours, the following generalization can be formed:

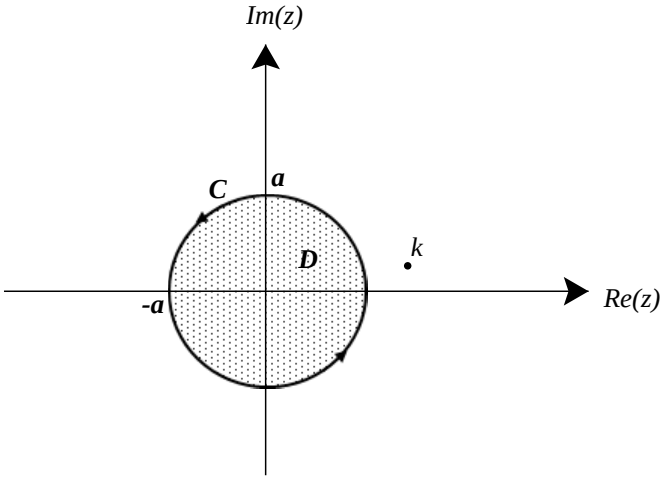
$$\oint_{C_0} f(z)\, dz = \sum_{p=1}^{n} \oint_{C_p} f(z) dz \qquad (*')$$

**SECTION 4.** EXAMPLES USING CAUCHY'S INTEGRAL THEOREM

**Example 1:** Suppose we are given the following complex function:

$$f(z) = \frac{z^2}{z-k} \qquad \text{where } z \neq k$$

We wish to evaluate the contour integral over the closed contour *C* described by $|z| = a$. First, we need to determine the domain of analyticity for *f(z)*. For this example, *f(z)* is analytic everywhere except when $z = k$. In order to apply the integral theorem, the closed contour can only encompass a regular domain of *f(z)*; therefore *k* cannot not be contained in the domain *D* (*D* is the enclosed region of contour *C*). Since $|z| = a$ is a circle of radius *a* centered at the origin; in order to use the integral theorem, the circle cannot encompass $z = k$ (therefore, $0 < a < |k|$). Now let us determine the integral:




$$\oint_C \frac{z^2}{z-k}\ dz$$

First we need to recall from elementary pre-calculus the method of long division of polynomials:

$$\begin{array}{r} z+k \\ z-k\overline{)\,z^2} \\ \underline{-z^2+kz} \\ kz \\ \underline{-kz+k^2} \\ k^2 \end{array}$$

Re-express the contour integral as the following:

$$\oint_C \frac{z^2}{z-k}\ dz = \oint_C z+k+\frac{k^2}{z-k}\ dz$$

$$= \oint_C z\ dz + \oint_C k\ dz + \oint_C \frac{k^2}{z-k}\ dz$$

$$\oint_C \frac{z^2}{z-k}\ dz = \frac{1}{2}z^2 + kz + k^2\log(z-k) + c$$

Evaluate this integral along the circle with radius = *a*. Convert *z* into polar format by making the following substitution:

$$z = ae^{i\theta}$$

$$\frac{1}{2}z^2 + kz + k^2\log(z-k) + c = \frac{1}{2}\left(ae^{i\theta}\right)^2 + kae^{i\theta} + k^2\log\left(ae^{i\theta}-k\right) + c$$

$$\oint \frac{z^2}{z-k}\ dz = \frac{1}{2}a^2e^{i2\theta} + kae^{i\theta} + k^2\log\left(ae^{i\theta}-k\right) + c$$

Now that we have the integral in polar form we can integrate over the circle from 0 radians to 2π radians.

$$\oint_C \frac{z^2}{z-k}\ dz = \frac{1}{2}a^2e^{i2\theta} + kae^{i\theta} + k^2\log\left(ae^{i\theta}-k\right)\Big|_{\theta=0}^{\theta=2\pi}$$

$$\oint_C \frac{z^2}{z-k}\ dz = \left(\frac{1}{2}a^2e^{i4\pi} + kae^{i2\pi} + k^2\log\left(ae^{i2\pi}-k\right)\right) - \left(\frac{1}{2}a^2e^{i2\cdot0} + kae^{i\cdot0} + k^2\log\left(ae^{i\cdot0}-k\right)\right)$$

$$\oint_C \frac{z^2}{z-k}\ dz = \left(\frac{1}{2}a^2 + ka + k^2\log(a-k)\right) - \left(\frac{1}{2}a^2 + ka + k^2\log(a-k)\right) = 0$$

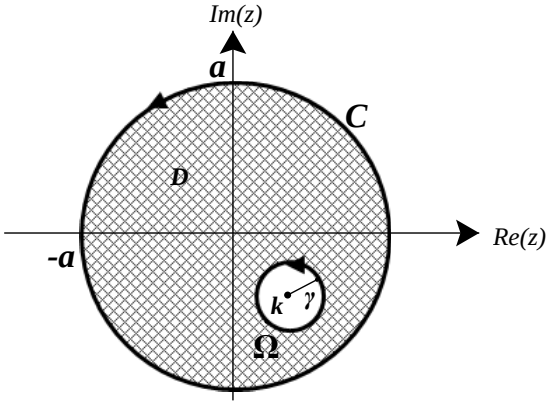
Therefore, $\oint_C \frac{z^2}{z-k}\ dz = 0$ over a contour *C* that does not include the singular point *z* = *k*. This agrees with Cauchy's Integral Theorem because the domain *D* is regular when, the radius of the contour circle *C*, $a < k$

**Example 2:** Using the same *f(z)* and contour *C* from Example 1, what does the integral evaluate to when the singularity *k* is located within the circle $|z| < a$?

Let us consider enclosing the singularity by another circular contour Ω, located entirely within the domain *D*, that had radius equal to γ. If we use section 3's conclusion regarding multiply-connected regions we can employ the following equation:

$$\oint_C f(z)\ dz = \oint_\Omega f(z)\ dz$$

We can describe the contour, Ω, encompassing the singularity *k*, in the following way:




$$|z-k|=\gamma$$

$$z-k=\gamma e^{i\theta} \qquad (1)$$

$$z=\gamma e^{i\theta}+k \qquad (2)$$

Substitute (1) and (2) into the integrand:

$$\oint \frac{z^2}{z-k}\,dz=\oint \frac{(\gamma e^{i\theta}+k)^2}{\gamma e^{i\theta}}\,\gamma i e^{i\theta}\,d\theta = i\oint\left(\gamma e^{i\theta}+k\right)^2 d\theta$$

$$i\oint \gamma^2 e^{i2\theta}+2\gamma k e^{i\theta}+k^2\,d\theta = i\left(-\frac{1}{2}i\gamma^2 e^{i2\theta}-2i\gamma k e^{i\theta}+k^2\theta\right)+c$$

$$\oint \frac{z^2}{z-k}\,dz=\frac{1}{2}\gamma^2 e^{i2\theta}+2\gamma k e^{i\theta}+ik^2\theta+c$$

Evaluate the above integral over the interval $0 \le \theta \le 2\pi$.

$$\oint_\Omega \frac{z^2}{z-k}\,dz=\frac{1}{2}\gamma^2 e^{i2\theta}+2\gamma k e^{i\theta}+ik^2\theta\Big|_0^{2\pi}=\left(\frac{1}{2}\gamma^2 e^{i4\pi}+2\gamma k e^{i2\pi}+ik^2 2\pi\right)-\left(\frac{1}{2}\gamma^2 e^{i0}+2\gamma k e^{i0}+ik^2 0\right)$$

$$\oint_\Omega \frac{z^2}{z-k}\,dz=\left(\frac{1}{2}\gamma^2+2\gamma k+i2\pi k^2\right)-\left(\frac{1}{2}\gamma^2+2\gamma k\right)=i2\pi k^2$$

$$\oint_C \frac{z^2}{z-k}\,dz=\oint_\Omega \frac{z^2}{z-k}\,dz=i2\pi k^2\neq 0$$

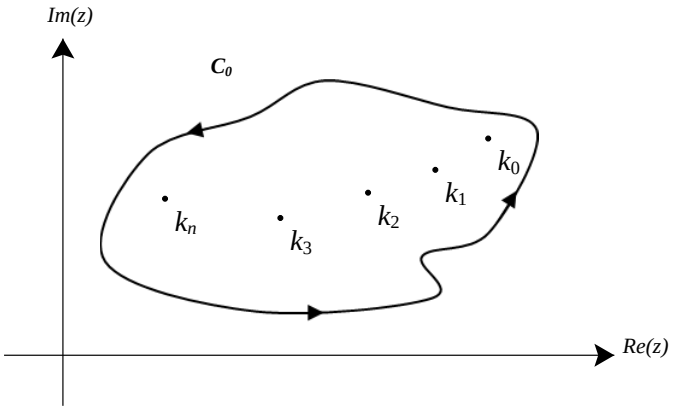
The contour integral along Ω does not equal 0. This makes sense because if the contour did not enclose the singularity *k* then the contour integral would equal 0 (hence we would use Cauchy's Integral Theorem).

**Example 3:** Given the following function:

$$f(z)=\frac{1}{P_n(z)}$$

Where $P_n(z)$ is a polynomial of degree n:

$$P_n(z)=a_n z^n+a_{n-1}z^{n-1}+\cdots+a_3 z^3+a_2 z^2+a_1 z+a_0$$
$$=\sum_{p=0}^{n} a_p z^p$$




For our purposes we will make the lead coefficient equal to 1, $a_n=1$. Determine the value of…

$$\oint_{C_0} f(z)\,dz$$

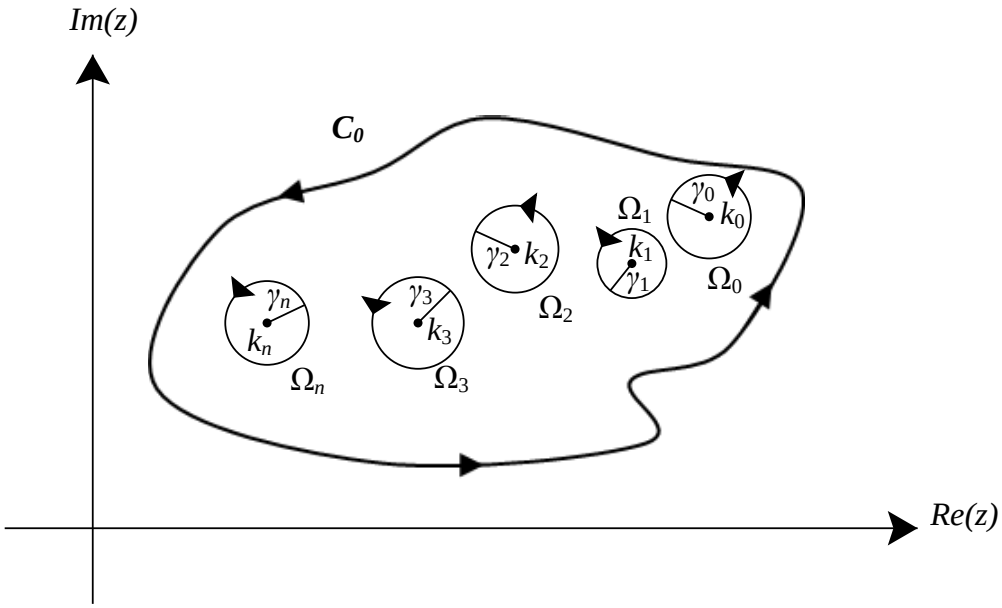
Where $C_0$ encompasses every pole of *f(z)*, please see the figure to the right. The poles will be equal to the zeros of $P_n(z)$, we will refer to them as $k_0, k_1, k_2, \ldots, k_n$. Each $k_p$ is distinct, there are no repeated roots—each pole is simple.

First, let us write $f(z)$ in terms of its poles, $k_0, k_1, k_2, \dots, k_n$, we will use partial fraction expansion methods:

$$f(z) = \frac{1}{P_n(z)} = \frac{A_0}{z-k_0} + \frac{A_1}{z-k_1} + \frac{A_2}{z-k_2} + \frac{A_3}{z-k_3} + \cdots \frac{A_n}{z-k_n}$$

$$f(z) = \frac{1}{P_n(z)} = \sum_{p=0}^{n} \frac{A_p}{z-k_p}$$

In order to determine the value of the integral $\oint_{C_0} f(z)\, dz$ we must draw a closed path around each pole. We cannot simply use Cauchy's Integral Theorem, because *f(z)* is not regular in the domain that $C_0$ encompasses.




Each closed path Ω can be described in the following way:

$$\Omega_n: \quad |z - k_n| = \gamma_n$$

$$z = \gamma_n e^{i\theta} + k_n$$

Using section 3 conclusion on multiply connected regions, we will utilize equation (*'), the integral on the larger closed path is equal to the sum of the integrals of interior closed paths.

$$\oint_{C_0} f(z)\, dz = \sum_{p=0}^{n} \oint_{\Omega_p} f(z) dz$$

Recall, that *f(z)* was expressed as a finite sum of partial fractions:

$$\oint_{C_0} f(z)\, dz = \sum_{p=0}^{n} \left[ \oint_{\Omega_p} \left( \sum_{j=0}^{n} \frac{A_j}{z-k_j} \right) dz \right]$$

Let us look at the problem for the first iteration of Ω:

$$\oint_{\Omega_0} \left( \sum_{j=0}^{n} \frac{A_j}{z-k_j} \right) dz = \oint_{\Omega_0} \left( \frac{A_0}{z-k_0} + \frac{A_1}{z-k_1} + \frac{A_2}{z-k_2} + \frac{A_3}{z-k_3} + \cdots \frac{A_n}{z-k_n} \right) dz$$

We shall make the substitution: $z = \gamma_0 e^{i\theta} + k_0$ and $dz = \gamma_0 i e^{i\theta}\, d\theta$

$$\oint_{\Omega_0} \left(\frac{A_0}{\gamma_0 e^{i\theta} + k_0 - k_0} + \frac{A_1}{\gamma_0 e^{i\theta} + k_0 - k_1} + \frac{A_2}{\gamma_0 e^{i\theta} + k_0 - k_2} + \cdots \frac{A_n}{\gamma_0 e^{i\theta} + k_0 - k_n}\right) \gamma_0 i e^{i\theta}\, d\theta$$

$$= \oint_{\Omega_0} \left(\frac{A_0 \gamma_0 i e^{i\theta}}{\gamma_0 e^{i\theta}} + \frac{A_1 \gamma_0 i e^{i\theta}}{\gamma_0 e^{i\theta} + k_0 - k_1} + \frac{A_2 \gamma_0 i e^{i\theta}}{\gamma_0 e^{i\theta} + k_0 - k_2} + \cdots \frac{A_n \gamma_0 i e^{i\theta}}{\gamma_0 e^{i\theta} + k_0 - k_n}\right) d\theta$$

$$= \oint_{\Omega_0} \left(i A_0 + \frac{A_1 \gamma_0 i e^{i\theta}}{\gamma_0 e^{i\theta} + k_0 - k_1} + \frac{A_2 \gamma_0 i e^{i\theta}}{\gamma_0 e^{i\theta} + k_0 - k_2} + \cdots \frac{A_n \gamma_0 i e^{i\theta}}{\gamma_0 e^{i\theta} + k_0 - k_n}\right) d\theta$$

The integral of the first term is trivial: $iA_0\theta$, the remaining require using some u-substitution:

$$u_m = \gamma_0 e^{i\theta} + k_0 - k_m$$

$$du = \gamma_0 i e^{i\theta}\, d\theta$$

Now we have from substitution:

$$\oint_{\Omega_0} \left(\sum_{j=0}^{n} \frac{A_j}{z - k_j}\right) dz = iA_0\theta + \oint_{\Omega_0} \left(\frac{A_1\, du}{u_1} + \frac{A_2\, du}{u_2} + \frac{A_3\, du}{u_3} + \cdots \frac{A_n\, du}{u_n}\right) = iA_0\theta + \oint_{\Omega_0} \left(\sum_{m=1}^{n} \frac{A_m}{u_m}\right) du$$

$$\oint_{\Omega_0} \left(\sum_{j=0}^{n} \frac{A_j}{z - k_j}\right) dz = iA_0\theta + \sum_{m=1}^{n} A_m \log(u_m) + c$$

$$\oint_{\Omega_0} \left(\sum_{j=0}^{n} \frac{A_j}{z - k_j}\right) dz = iA_0\theta + \sum_{m=1}^{n} A_m \log\left(\gamma_0 e^{i\theta} + k_0 - k_m\right) + c$$

Evaluate over a full circle $0 < \theta < 2\pi$:

$$\left[iA_0 2\pi + \sum_{m=1}^{n} A_m \log\left(\gamma_0 e^{i2\pi} + k_0 - k_m\right)\right] - \left[iA_0 0 + \sum_{m=1}^{n} A_m \log\left(\gamma_0 e^{i0} + k_0 - k_m\right)\right]$$

The summation terms will cancel each other out entirely and you are left with:

$$\oint_{\Omega_0} \left(\sum_{j=0}^{n} \frac{A_j}{z - k_j}\right) dz = iA_0 2\pi$$

If we repeat this calculation for the next interior sub contours we obtain the following:

$$\oint_{\Omega_1} \left(\sum_{j=0}^{n} \frac{A_j}{z - k_j}\right) dz = iA_1 2\pi$$

$$\oint_{\Omega_2} \left(\sum_{j=0}^{n} \frac{A_j}{z - k_j}\right) dz = iA_2 2\pi$$

…

$$\oint_{\Omega_n} \left(\sum_{j=0}^{n} \frac{A_j}{z - k_j}\right) dz = iA_n 2\pi$$

The integral over all of $C_0$ is equal to the sum of the each one of the sub contour integrals:

$$\oint_{C_0} f(z)\,dz = \sum_{p=0}^{n} \oint_{\Omega_p} f(z)dz = \sum_{p=0}^{n} iA_p 2\pi$$

$$\oint_{C_0} f(z)\,dz = 2\pi i(A_0 + A_1 + A_2 + \cdots A_n)$$

Hence the integral along a closed contour encompassing all the poles of *f(z)* is equal to the product of $2\pi i$ and the sum of the partial fraction numerators.

**Example 4:** Given the following function:

$$f(z) = \frac{g(z)}{z-k}$$

Let us determine the following integral, where *C* contains the pole *k*:

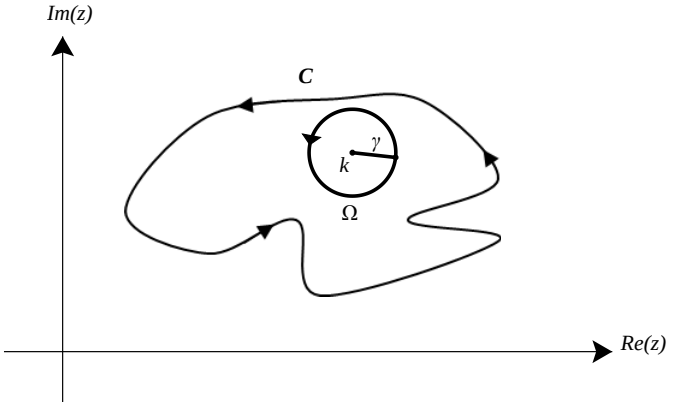
$$\oint_C f(z)\,dz$$

Again, since the contour contains a simple pole, we will draw another contour called Ω around *k*, with radius γ. We will describe Ω the following way:

$$|z-k| = \gamma$$

$$z = \gamma e^{i\theta} + k$$

$$dz = \gamma i e^{i\theta}\,d\theta$$

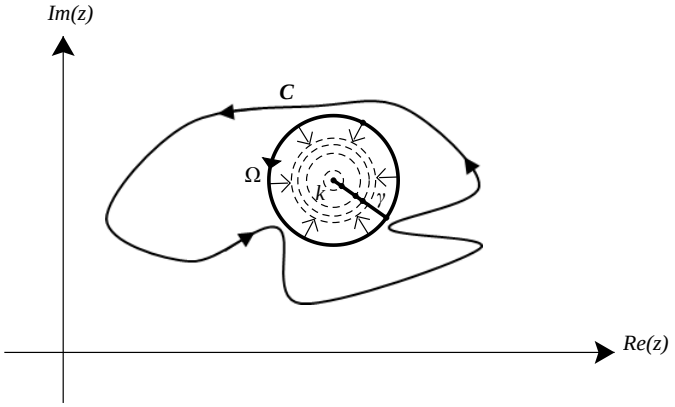



Using section 4 conclusions we can use the following relationship:

$$\oint_C f(z)\,dz = \oint_\Omega f(z)\,dz$$

$$\oint_\Omega f(z)\,dz = \oint_\Omega \frac{g(z)}{z-k}\,dz = \int_0^{2\pi} \frac{g\left(\gamma e^{i\theta}+k\right)}{\gamma e^{i\theta}} \gamma i e^{i\theta}\,d\theta$$

$$\oint_\Omega f(z)\,dz = i\int_0^{2\pi} g\left(\gamma e^{i\theta}+k\right)d\theta$$




If we take the limit on the right side of the equal sign as $\gamma \to 0$, because we want to shrink the Ω contour infinitely small.

$$\oint_\Omega f(z)\,dz = \lim_{\gamma\to 0} i\int_0^{2\pi} g\left(\gamma e^{i\theta}+k\right)d\theta = i\int_0^{2\pi} \lim_{\gamma\to 0} g\left(\gamma e^{i\theta}+k\right)\,d\theta$$

$$\oint_\Omega f(z)\,dz = \; i\int_0^{2\pi} g(k)\,d\theta = ig(k)\theta|_0^{2\pi} = ig(k)(2\pi) - ig(k)(0) = i2\pi g(k)$$

Therefore,

$$\oint_C f(z)\,dz = \oint_\Omega \frac{g(z)}{z-k}\,dz = \; i2\pi g(k)$$

**Example 5:** Suppose we are given the following function.

$$f(z) = \frac{g(z)}{P_n(z)}$$

$P_n(z)$ is a polynomial of degree n, and the lead coefficient is equal to 1. With this in mind we can use partial fraction expansions to express the rational portion of *f(z)*.

$$f(z) = \frac{g(z)}{P_n(z)} = g(z)\left(\frac{A_0}{z-k_0} + \frac{A_1}{z-k_1} + \frac{A_2}{z-k_2} + \frac{A_3}{z-k_3} + \cdots \frac{A_n}{z-k_n}\right)$$

$$f(z) = \frac{g(z)}{P_n(z)} = g(z)\sum_{p=0}^{n}\frac{A_p}{z-k_p}$$

If we integrate over a contour that does not contain any singularities then we can apply Cauchy's Integral Theorem. For this example it will be more interesting to find out what the integral will evaluate to when the contour encompasses all of the poles. First, we will restrict the function *g(z)* such that it is analytic for all of the *z*-plane, and each *k* will be distinct. We will employ the same kind of logic as was shown in example 3, we will imagine encompassing each singularity *k* with its own circular contour Ω. Then, using the integral theorem over multiply connected domains, (see section 3), we obtain the following integral:

$$\oint_C f(z)\,dz = \sum_{p=0}^{n}\left[\oint_{\Omega_p}\left(g(z)\sum_{j=0}^{n}\frac{A_j}{z-k_j}\right)dz\right]$$

Where, Ω is defined as

$$\Omega_n: \quad |z-k_n| = \gamma_n$$

$$z = \gamma_n e^{i\theta} + k_n$$

Consider the first iteration of Ω for n = 0:

$$\oint_{\Omega_0}\left(g(z)\sum_{j=0}^{n}\frac{A_j}{z-k_j}\right)dz = \oint_{\Omega_0} g(z)\left(\frac{A_0}{z-k_0} + \frac{A_1}{z-k_1} + \frac{A_2}{z-k_2} + \frac{A_3}{z-k_3} + \cdots \frac{A_n}{z-k_n}\right)dz$$

We shall make the substitution: $z = \gamma_0 e^{i\theta} + k_0$ and $dz = \gamma_0 i e^{i\theta}\,d\theta$

$$\oint_{\Omega_0} g(\gamma_0 e^{i\theta}+k_0)\left(\frac{A_0}{\gamma_0 e^{i\theta}+k_0-k_0} + \frac{A_1}{\gamma_0 e^{i\theta}+k_0-k_1} + \frac{A_2}{\gamma_0 e^{i\theta}+k_0-k_2} + \cdots \frac{A_n}{\gamma_0 e^{i\theta}+k_0-k_n}\right)\gamma_0 i e^{i\theta}\,d\theta$$

$$= \oint_{\Omega_0} g(\gamma_0 e^{i\theta}+k_0)\left(\frac{A_0\gamma_0 i e^{i\theta}}{\gamma_0 e^{i\theta}} + \frac{A_1\gamma_0 i e^{i\theta}}{\gamma_0 e^{i\theta}+k_0-k_1} + \frac{A_2\gamma_0 i e^{i\theta}}{\gamma_0 e^{i\theta}+k_0-k_2} + \cdots \frac{A_n\gamma_0 i e^{i\theta}}{\gamma_0 e^{i\theta}+k_0-k_n}\right)d\theta$$

$$= \oint_{\Omega_0} g(\gamma_0 e^{i\theta}+k_0)\left(iA_0 + \frac{A_1\gamma_0 i e^{i\theta}}{\gamma_0 e^{i\theta}+k_0-k_1} + \frac{A_2\gamma_0 i e^{i\theta}}{\gamma_0 e^{i\theta}+k_0-k_2} + \cdots \frac{A_n\gamma_0 i e^{i\theta}}{\gamma_0 e^{i\theta}+k_0-k_n}\right)d\theta$$

If we take the limit in the above line and let $\gamma_0 \to 0$, then all of the terms with the exception of the trivial term $iA_0$ cancel out. When we take the limit imagine that the circles around each pole shrink closer and closer to their respective centers *k*.

$$\oint_{\Omega_0}\left(g(z)\sum_{j=0}^{n}\frac{A_j}{z-k_j}\right)dz = \oint_{\Omega_0} g(k_0)(iA_0)\,d\theta = \int_0^{2\pi} g(k_0)(iA_0)\,d\theta = i2\pi A_0 g(k_0)$$

$$\oint_{\Omega_0} \left( g(z) \sum_{j=0}^{n} \frac{A_j}{z - k_j} \right) dz = \; i2\pi A_0 g(k_0)$$

We can repeat these steps for each contour around a singularity and obtain the following:

$$\oint_{\Omega_1} \left( g(z) \sum_{j=0}^{n} \frac{A_j}{z - k_j} \right) dz = \; i2\pi A_1 g(k_1)$$

$$\oint_{\Omega_2} \left( g(z) \sum_{j=0}^{n} \frac{A_j}{z - k_j} \right) dz = \; i2\pi A_2 g(k_2)$$

……

$$\oint_{\Omega_n} \left( g(z) \sum_{j=0}^{n} \frac{A_j}{z - k_j} \right) dz = \; i2\pi A_n g(k_n)$$

Replace these quantities into the original integral equation:

$$\oint_C f(z)\, dz = \sum_{p=0}^{n} \left[ \oint_{\Omega_p} \left( g(z) \sum_{j=0}^{n} \frac{A_j}{z - k_j} \right) dz \right] = \sum_{p=0}^{n} i2\pi A_p g(k_p)$$

Therefore,

$$\oint_C \frac{g(z)}{P_n(z)} dz = i2\pi \big( A_0 g(k_0) + A_1 g(k_1) + \cdots A_n g(k_n) \big)$$

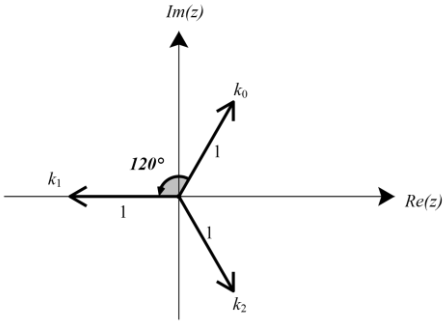
**Example 6:** Given the function

$$f(z) = \frac{1}{1 + z^3}$$

Let us determine the integral over a contour encompassing one of the singularities. First, we will determine the singular points:

$$1 + z^3 = 0 \quad \leftrightarrow \quad z^3 = -1 = e^{i(2n+1)\pi}$$

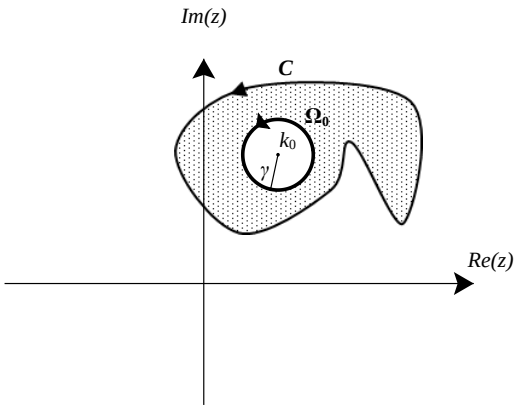
$$z = e^{i(2n+1)\frac{\pi}{3}} \quad \text{where} \quad n = 0, 1, 2 \;\rightarrow\; \text{singular values } k = \{e^{i\frac{\pi}{3}}, e^{i\pi}, e^{i\frac{5\pi}{3}}\}$$




To the right is an argand diagram indicating the zeros to the denominator of *f(z)*-these are also the singularities of *f(z)*. We wish to evaluate the integral along a closed contour encompassing one of the *k* values (singularities). Let us choose $k_0 = e^{i\frac{\pi}{3}}$. We cannot use Cauchy's Integral Theorem to evaluate the integral because *f(z)* is not regular inside the contour *C*; therefore, enclose the singularity with a new contour. We will name this contour $\Omega_0$.

$$\oint_C f(z)\, dz = \oint_{\Omega_0} f(z)\, dz$$

Integration about $\Omega_0$ by using the following substitution:




$$\Omega_0: \quad z = \gamma e^{i\theta} + k_0$$

$$dz = \gamma i e^{i\theta}\, d\theta$$

Substitute the variable and the differential and obtain the following:

$$\oint_{\Omega_0} f(z)\, dz = \int \frac{dz}{1+z^3} = \int \frac{\gamma i e^{i\theta}\, d\theta}{1+(\gamma e^{i\theta}+k_0)^3}$$

We need to expand the cubic term:

$$\left(\gamma e^{i\theta}+k_0\right)^3 = \left(\gamma e^{i\theta}+k_0\right)\left(\gamma e^{i\theta}+k_0\right)\left(\gamma e^{i\theta}+k_0\right) = \left(\gamma^2 e^{i2\theta}+2k_0\gamma e^{i\theta}+k_0^3\right)\left(\gamma e^{i\theta}+k_0\right)$$

$$\left(\gamma e^{i\theta}+k_0\right)^3 = \left(\gamma^3 e^{i3\theta}+2k_0\gamma^2 e^{i2\theta}+k_0^2\gamma e^{i\theta}+k_0\gamma^2 e^{i2\theta}+2k_0^2\gamma e^{i\theta}+k_0^3\right) \quad \text{remember } k_0^3 = -1$$

$$\left(\gamma e^{i\theta}+k_0\right)^3 = \gamma^3 e^{i3\theta}+3k_0\gamma^2 e^{i2\theta}+3k_0^2\gamma e^{i\theta}-1$$

Therefore,

$$\int \frac{dz}{1+z^3} = \int \frac{\gamma i e^{i\theta}\, d\theta}{1+(\gamma e^{i\theta}+k_0)^3} = \int \frac{\gamma i e^{i\theta}\, d\theta}{\gamma^3 e^{i3\theta}+3k_0\gamma^2 e^{i2\theta}+3k_0^2\gamma e^{i\theta}}$$

Next we can factor out $\gamma e^{i\theta}$:

$$\int \frac{\gamma i e^{i\theta}\, d\theta}{\gamma e^{i\theta}(\gamma^2 e^{i2\theta}+3k_0\gamma e^{i\theta}+3k_0^2)} = \int \frac{i\, d\theta}{\gamma^2 e^{i2\theta}+3k_0\gamma e^{i\theta}+3k_0^2}$$

As in the previous example, let us take the limit as $\gamma \to 0$, this will cancel out the first two terms in the denominator of the above integrand. Remember, imagine that the circle $\Omega_0$ is shrinking closer and closer to the singularity.

$$\lim_{\gamma\to 0}\int \frac{i\, d\theta}{\gamma^2 e^{i2\theta}+3k_0\gamma e^{i\theta}+3k_0^2} = \int \lim_{\gamma\to 0}\frac{i\, d\theta}{\gamma^2 e^{i2\theta}+3k_0\gamma e^{i\theta}+3k_0^2} = \int \frac{i\, d\theta}{3k_0^2} = \frac{i\theta}{3k_0^2}+c$$

We wish to evaluate over the interval $0 \le \theta \le 2\pi$.

$$\oint_{\Omega_0} f(z)\, dz = \frac{i2\pi}{3k_0^2}$$

Let us remember that $k_0 = e^{i\frac{\pi}{3}}$:

$$\oint_{\Omega_0} f(z)\, dz = \frac{i2\pi}{3k_0^2} = \frac{i2\pi}{3e^{i\frac{2\pi}{3}}} = i\frac{2}{3}\pi e^{-i\frac{2}{3}\pi} = i\frac{2}{3}\pi\left(-\frac{1}{2}-i\frac{\sqrt{3}}{2}\right)$$

$$\oint_{\Omega_0} f(z)\, dz = \pi\left(\frac{\sqrt{3}}{3}-i\frac{1}{3}\right)$$

$$\oint_{C} f(z)\, dz = \pi\left(\frac{\sqrt{3}}{3}-i\frac{1}{3}\right)$$

As a corollary to this problem, if we consider encompassing all three of the singularities then we would have to encompass each singularity with its own circular contour Ω, each time we will obtain the same result as shown above therefore if *C* includes all three singularities then the following is true:

$$\oint_C f(z) = \frac{i2\pi}{3}\left(\frac{1}{k_0^2}+\frac{1}{k_1^2}+\frac{1}{k_2^2}\right)$$

**Example 7:** For this example we shall consider the general case of example 6 where *p* is any positive integer greater than 0, we will evaluate the contour integral enclosing one of the singularities and then we will generalize this for all the singularities.

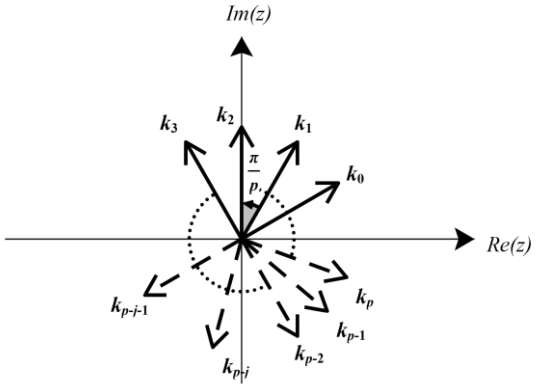
$$f(z) = \frac{1}{1+z^p}$$

But we must first recall the binomial theorem [6]:

$$(a+b)^p = \sum_{r=0}^{p}\binom{p}{r}a^{p-r}b^r \quad \text{where } \binom{p}{r} = \frac{p!}{r!\,(p-r)!}$$

As in the previous example, let us determine the zeros of the denominator:

$$z^p = -1 = e^{i(2n+1)\pi} \quad \leftrightarrow \quad z = e^{i(2n+1)\frac{\pi}{p}} \quad n = \{0,1,2,3\ldots,p-1\}$$




These solutions imply we will have *p* vectors in an argand diagram that are *π/p* radians apart about the origin. Please see the figure at the right. Now imagine if we wish to calculate the integral along a contour encompassing just one of these singularities, we would have to evaluate the integral of a circular contour that encloses our singularity. Let us use $k_0$ and enclose it with a circular path $\Omega_0$.

$$\oint_C f(z)\,dz = \oint_{\Omega_0} f(z)\,dz$$

We will make a similar substitution as we did in example 6, keep in mind $k_0$ is the solution $e^{i\frac{\pi}{p}}$:

$$z = \gamma e^{i\theta} + k_0$$

$$dz = \gamma i e^{i\theta}$$

If we make the replacement into the denominator of *f(z)* we obtain the following:

$$f(z) = \frac{1}{1+(\gamma e^{i\theta}+k_0)^p}$$

This is where we need to use the binominal theorem and expand the bottom power:

$$\left(\gamma e^{i\theta}+k_0\right)^p = \sum_{r=0}^{p}\binom{p}{r}\left(\gamma e^{i\theta}\right)^{p-r}k_0^r$$

$$\sum_{r=0}^{p}\binom{p}{r}\left(\gamma e^{i\theta}\right)^{p-r}{k_0}^r = \gamma^p e^{ip\theta} + \binom{p}{1}\gamma^{p-1}e^{i(p-1)\theta}k_0 + \binom{p}{2}\gamma^{p-2}e^{i(p-2)\theta}k_0^2 + \binom{p}{3}\gamma^{p-3}e^{i(p-3)\theta}k_0^3 + \cdots + \binom{p}{p}k_0^p$$

The last term $\binom{p}{p} k_0^p$ will evaluate to -1; therefore,

$$\sum_{r=0}^{p} \binom{p}{r} \left(\gamma e^{i\theta}\right)^{p-r} {k_0}^r = \gamma^p e^{ip\theta} + \binom{p}{1} \gamma^{p-1} e^{i(p-1)\theta} k_0 + \binom{p}{2} \gamma^{p-2} e^{i(p-2)\theta} k_0^2 + \binom{p}{3} \gamma^{p-3} e^{i(p-3)\theta} k_0^3 + \cdots + \binom{p}{p-1} \gamma e^{i\theta} k_0^{p-1} - 1$$

We can now replace this into the denominator of our function *f(z)*:

$$f(z) = \frac{1}{1 + \gamma^p e^{ip\theta} + \binom{p}{1} \gamma^{p-1} e^{i(p-1)\theta} k_0 + \binom{p}{2} \gamma^{p-2} e^{i(p-2)\theta} k_0^2 + \binom{p}{3} \gamma^{p-3} e^{i(p-3)\theta} k_0^3 + \cdots + \binom{p}{p-1} \gamma e^{i\theta} k_0^{p-1} - 1}$$

$$f(z) = \frac{1}{\gamma^p e^{ip\theta} + \binom{p}{1} \gamma^{p-1} e^{i(p-1)\theta} k_0 + \binom{p}{2} \gamma^{p-2} e^{i(p-2)\theta} k_0^2 + \binom{p}{3} \gamma^{p-3} e^{i(p-3)\theta} k_0^3 + \cdots + \binom{p}{p-1} \gamma e^{i\theta} k_0^{p-1}}$$

If we recall from the previous example, we had to factor out $\gamma e^{i\theta}$:

$$f(z) = \frac{1}{\gamma e^{i\theta} \left(\gamma^{p-1} e^{i(p-1)\theta} + \binom{p}{1} \gamma^{p-2} e^{i(p-2)\theta} k_0 + \binom{p}{2} \gamma^{p-3} e^{i(p-3)\theta} k_0^2 + \binom{p}{3} \gamma^{p-4} e^{i(p-4)\theta} k_0^3 + \cdots + \binom{p}{p-1} k_0^{p-1}\right)}$$

Now if we include the differential *dz*:

$$f(z)\, dz = \frac{\gamma i e^{i\theta} d\theta}{\gamma e^{i\theta} \left(\gamma^{p-1} e^{i(p-1)\theta} + \binom{p}{1} \gamma^{p-2} e^{i(p-2)\theta} k_0 + \binom{p}{2} \gamma^{p-3} e^{i(p-3)\theta} k_0^2 + \binom{p}{3} \gamma^{p-4} e^{i(p-4)\theta} k_0^3 + \cdots + \binom{p}{p-1} k_0^{p-1}\right)}$$

We can cancel out $\gamma e^{i\theta}$:

$$f(z)dz = \frac{id\theta}{\gamma^{p-1} e^{i(p-1)\theta} + \binom{p}{1} \gamma^{p-2} e^{i(p-2)\theta} k_0 + \binom{p}{2} \gamma^{p-3} e^{i(p-3)\theta} k_0^2 + \binom{p}{3} \gamma^{p-4} e^{i(p-4)\theta} k_0^3 + \cdots + \binom{p}{p-1} k_0^{p-1}}$$

If we take the limit as $\gamma \to 0$ all the bottom terms cancel out with the exception of the last:

$$f(z)dz = \frac{id\theta}{\binom{p}{p-1} k_0^{p-1}}$$

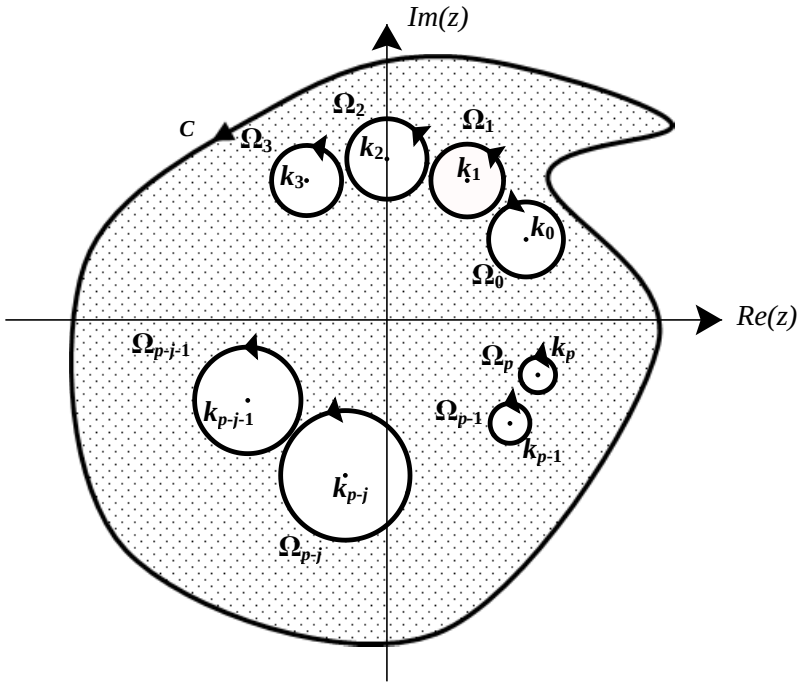
Take the integral over the interval $0 \le \theta \le 2\pi$.

$$\oint_{\Omega_0} f(z)dz = \int_0^{2\pi} \frac{id\theta}{\binom{p}{p-1} k_0^{p-1}} = \frac{2\pi i}{\binom{p}{p-1} k_0^{p-1}}$$

Hence,

$$\oint_C f(z)\, dz = \frac{2\pi i}{\binom{p}{p-1} k_0^{p-1}}$$

Now if *C* encloses all of the singularities the integral over the whole contour will be the sum of the contour integrals of every Ω contour:




$$\oint_C f(z)\,dz = \oint_{\Omega_0} f(z)\,dz + \oint_{\Omega_1} f(z)\,dz + \oint_{\Omega_2} f(z)\,dz + \cdots + \oint_{\Omega_p} f(z)\,dz$$

Through substitution, and simplification we arrive at the following statement.

$$\oint_C f(z)\,dz = \frac{2\pi i}{\binom{p}{p-1} k_0^{p-1}} + \frac{2\pi i}{\binom{p}{p-1} k_1^{p-1}} + \frac{2\pi i}{\binom{p}{p-1} k_1^{p-1}} + \cdots + \frac{2\pi i}{\binom{p}{p-1} k_p^{p-1}}$$

$$\oint_C f(z)\,dz = \frac{2\pi i}{\binom{p}{p-1}} \sum_{j=0}^{p} \frac{1}{k_j^{p-1}}$$

**CONCLUDING REMARKS:** Through this paper the following was demonstrated: the value of a contour integral between two points is independent of the path taken, singularities contained in a closed path affect the value of the contour integral along that path, and if a contour encloses many sub contours then the integral along the exterior contour is equal to the sum of the integrals about each sub contour. This is relevant because many questions arise regarding line integrals, path independence, and even in physical applications regarding energy and the paths taken. Secondly, how do singular values affect the value of the integral along a path; for example, imagine moving from point A to point B, but there is a black hole, how does this singularity affect our path from A to B. More questions arise from this paper; for instance, in the last example does the final statement still hold if p = ½? Another question that can be ventured is how do we integrate if the singularity is right on the path *C* we have chosen? Most of the functions used in this paper were elementary functions, what about non-elementary functions, do their contour integrals follow through and have a multiplicity of $2\pi i$?

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